Demodulate, Then Attend: Exposing Amplitude Dynamics for Compact EEG Models

Abstract

Large pretrained models are becoming central to electroencephalography (EEG) analysis, yet compact models trained from scratch can remain competitive. We examine this observation through the signal representation presented to the encoder. Clinical EEG contains both waveform morphology and changes in the amplitude of frequency-band activity. These changes are present in a raw waveform, but their nonlinear extraction remains part of the learning problem when tokens are formed by linear patch embedding. We present demodulation-first tokenization: extract bandwise amplitude envelopes before patch embedding, preserve their temporal structure, and retain band-limited waveforms as complementary content. Analytic demodulation and scattering theory motivate this separation between rapid oscillations and amplitude dynamics. A compact Transformer processes the resulting electrode–band–time grid. A reference model with 1.62 million parameters achieves Cohen's $\kappa = 0.711$ on TUEV when trained from scratch, without cross-band phase rotation. Component comparisons further reveal different contributions of waveform content to event classification and seizure detection. Our work investigates how exposing useful signal structure before attention can make compact EEG models effective, and which temporal details a tokenizer should preserve for the encoder to learn.

1 Introduction

EEG representation learning has progressed from compact convolutional networks to Transformers pretrained on large collections of recordings [3, 4, 10, 12]. This progression promises increasingly transferable representations for clinical interpretation. Yet model scale does not consistently determine downstream performance: recent benchmarking finds that supervised specialists remain competitive with pretrained EEG models across many tasks [6]. Our own compact models motivate the same question: which properties of the input allow a small model to compete with a larger pretrained encoder? We investigate the representation on which the encoder operates, and the signal computations that representation makes explicit.

A clinical EEG event has structure at more than one temporal scale. Individual discharges have a shape, duration, and polarity; repeated activity also has an onset, persistence, and pattern of recurrence. Clinical terminology describes morphology, frequency, temporal organization, and spatial distribution [2]. These properties motivate complementary views of a recording. Band-limited waveforms retain local oscillatory shape, while their amplitude envelopes describe how activity strengthens and weakens over time. Exposing both views provides a way to represent a transient's shape together with the activity pattern in which it occurs.

Raw-patch tokenization does not explicitly make this separation. A linear projection maps samples within a patch into an embedding, leaving the encoder to learn useful nonlinear summaries. An amplitude envelope is one such summary: it requires a band-selective transformation followed by a nonlinear modulus. A linear patch embedding cannot perform this computation by itself. Consequently, preserving the waveform and exposing its envelope are different computational choices, even when both originate from exactly the same samples. We hypothesize that making the latter available at the input can reduce the feature-extraction burden on a compact encoder.

Analytic demodulation provides a signal-processing basis for this hypothesis. The modulus of a band's analytic signal yields its amplitude envelope; for a narrowband amplitude-modulated component, this exposes the slower amplitude variation carried by the oscillation. Scattering theory develops this principle through cascaded bandpass transforms and modulus operations [1]. It also identifies a second issue: averaging the demodulated response removes temporal modulations that may remain discriminative. The implication for tokenization is specific. Extracting an envelope and preserving the structure of that envelope are separate design decisions.

Consider two intervals with similar average band energy. One contains sustained activity; the other contains several separated bursts. A single energy summary cannot specify this distinction, whereas an amplitude trajectory can. Conversely, two transients with similar envelopes can differ in waveform shape. This suggests a tokenizer that keeps a short trajectory within each band and a corresponding waveform view. Both should refer to the same electrode, band, and time interval, so that the encoder can learn their relationship without first inferring which observations belong together.

Existing time–frequency frontends implement parts of this idea in different ways. CBraMod adds a projected patch spectrum to convolutional waveform features [12]. TFM-Tokenizer learns and quantizes motifs from waveform and short-time spectral representations [8]. The Morlet Spectral Transformer constructs time-resolved amplitude features using wavelet modulus and temporal pooling [9]. Their success makes amplitude information a substantive precedent. It also

motivates a more specific question: how should amplitude dynamics and waveform detail be represented before they are compressed and mixed by the encoder? We examine this question through explicit band-content views and component comparisons with a fixed encoder within each comparison.

Our reference tokenizer first decomposes each channel with a learnable bandpass filterbank. An analytic transform extracts the amplitude envelope of each filtered signal. Shared projections embed short sequences of log-envelope samples and band-limited waveform samples into two sets of token coordinates. The envelope projection receives the sequence itself, rather than a single average over the patch. The resulting tokens retain an electrode–band–time grid, processed by a compact axial Transformer. In the unrotated reference, the two content views are supplied directly to the encoder. This construction separates the extraction of signal content from the subsequent learning of relationships among tokens.

The reference model reaches Cohen's $\kappa = 0.711$ on TUEV with 1.62 million parameters and no pretraining. Component tests in the same model family also show that the two content views make different contributions across tasks. In matched seed-2 comparisons with the phase rotation held fixed, removing waveform content changes TUEV κ from 0.715 to 0.714, but changes CHB-MIT PR-AUC from 0.710 to 0.543 under the same evaluation protocol. These observations support retaining complementary content while examining its contribution at the task level. They motivate our central hypothesis: directly exposing amplitude dynamics can simplify the representation problem faced by a small encoder, while waveform content preserves information useful for other clinical distinctions.

The focus of this work is the choice of signal content at the tokenizer–encoder interface. We bring demodulation, temporal detail, and waveform morphology into a common token construction, and use a compact reference model to study their predictive roles. More broadly, we ask which signal computations should be supplied explicitly before attention, so that learned capacity can be devoted to the structure that remains.

2 Related Work

2.1 EEG pretraining and tokenization

EEG foundation models seek transferable representations across subjects, tasks, and recording configurations. BIOT organizes channelwise signal segments into a common token sequence for cross-data learning [13]. LaBraM learns discrete neural codes through spectrum prediction and uses these codes as targets for masked pretraining [3]. EEGPT combines masked learning with representation alignment [11], while CBraMod separates spatial and temporal interactions in a criss-cross Transformer [12]. These approaches address transfer and structural heterogeneity through different input representations and learning objectives.

TFM-Tokenizer places the frontend at the center of this pipeline. Its spectral path models interactions among frequency patches, combines them with waveform features, and learns a discrete vocabulary of time–frequency motifs [8]. This demonstrates that token construction can support both compact downstream models and existing encoders. We study a complementary design choice: explicitly extracting continuous amplitude trajectories and retaining their corresponding waveforms in each band. The distinction concerns what enters a token and which operations precede the encoder, rather than whether a model uses frequency information at all.

2.2 Frequency-resolved and amplitude-based EEG representations

Explicit spectral structure has a long history in EEG decoding. Schirrmeister et al. incorporate squaring, temporal pooling, and a logarithm in a shallow convolutional architecture designed around band-power extraction [10]. EEGNet uses depthwise and separable convolutions to build compact temporal and spatial feature extractors [4]. FBCNet combines a filterbank, spatial filtering, and temporal variance aggregation [7]. These methods demonstrate that useful signal-processing operations can be built into small learned models.

Recent Transformer frontends extend this approach. CBraMod combines patchwise FFT magnitudes with waveform features [12]; the Morlet Spectral Transformer applies a fixed wavelet bank, takes its modulus, pools it over time, and learns frequency-specific spatial projections [9]. The latter already represents amplitude variation across time bins. Our construction instead projects a sequence of envelope samples within each token and pairs it with the waveform from the same band, while retaining electrodes as a token axis. These differences make the granularity of amplitude representation and the contribution of waveform detail concrete objects of study.

2.3 Demodulation and representations for compact models

Demodulation separates an oscillatory component from its amplitude variation. In the deep scattering spectrum, analytic wavelet transforms and modulus operations expose subband envelopes, and further transforms recover modulation structure lost through local averaging [1]. This provides a useful distinction between extracting amplitude and retaining its dynamics. Our tokenizer follows the former signal-processing principle and supplies the resulting trajectories to learned attention, alongside waveform content.

JiT distinguishes the network's direct prediction from the space in which its training loss is evaluated, showing that the prediction parameterization can determine whether an apparently under-capacity network succeeds [5]. EEG-FM-Compass likewise finds that larger pretrained EEG models do not consistently outperform supervised specialists [6]. Our question concerns the input side of this capacity problem: whether explicitly exposing discriminative signal structure lets a compact encoder use its learned capacity more effectively.

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